EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test 3, Fall 2025

Course: CSE 207 Data Structures, Section-06 **Date:** December 11, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor
Full Marks: 10
Time: 25 Minutes

Note: There are **TWO** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

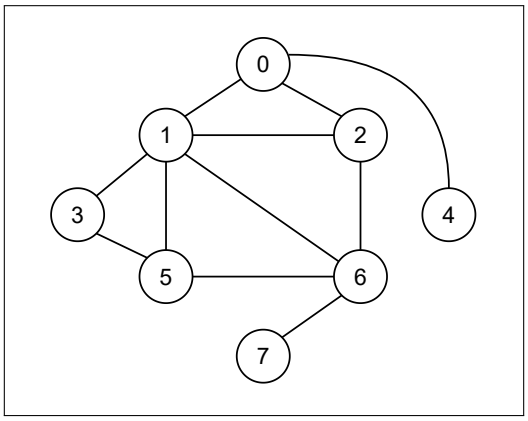


Figure 1: A Graph with 8 nodes

1. Given the graph in **Figure 1**:

(a) Find the **degree of each node**.

(b) Show the **adjacency list representation** of the given graph.

[CO2, C3, Mark: 5]

2. Perform **Breadth First Traversal (BFS)** of the graph in **Figure 1** to find the **minimum distance from source node 3** to all other nodes. You must show:

(a) The **BFS Queue status at each step**

(b) The **Parent node** information for each node

(c) The **Distance** of each node from the source

(d) The final **BFS visit order**

Instruction: You may **cross out outdated/removed values** in your data structures; there is no need to rewrite them repeatedly.

[CO2, C3, Mark: 5]